

Chapter 18 Calculations Worksheet

Edexcel A-Level Economics · Theme 3 · Revenue, costs and profit | Name: _____

Date: _____

Show the formula, your working and the answer with units. Answers are on the last page.

Section A: Short calculations

1. A firm sells 400 units at £15 each. Calculate total revenue. (2)
2. Total revenue is £300 when 10 units are sold and £330 when 11 units are sold. Calculate marginal revenue. (2)
3. A firm has total fixed costs of £2,000 and average variable cost of £5 when it produces 500 units. Calculate total cost and average cost. (3)
4. Total cost rises from £1,200 at 100 units to £1,215 at 101 units. Calculate marginal cost. (2)
5. A firm sells 800 units at £12 each. Its total cost is £8,200. Calculate its profit and its profit margin to 1 d.p. (4)
6. The price of a product rises from £8 to £10 and quantity demanded falls from 500 to 425. Calculate PED, and state the effect on total revenue. (4)
7. Profit falls from £50,000 to £42,000. Calculate the percentage change in profit. (2)
8. A firm's AR is £6, its AVC is £5.50 and its AC is £7. Should it continue producing in the short run? Should it stay in the market in the long run? Explain. (4)

Section B: Complete the table

A firm faces the following demand and cost figures.

Output (Q)	Price = AR (£)	TR (£)	MR (£)	TC (£)	MC (£)	AC (£)	Profit (£)
1	20			12			
2	18			22			
3	16			30			
4	14			40			
5	12			54			
6	10			72			

9. State the profit-maximising output and the maximum profit. Explain using MC and MR. (4)
10. State the output at which revenue is maximised. Explain your answer using MR. (3)
11. State the highest output at which the firm still earns at least normal profit ($AR \geq AC$). Which business objective does this represent? (3)
12. Compare the price charged under profit maximisation with the price under sales maximisation. (2)

Answers

Chapter 18 Calculations Worksheet

1. $TR = P \times Q = 15 \times 400 = \text{£}6,000$
2. $MR = \text{change in } TR \div \text{change in } Q = (330 - 300) \div 1 = \text{£}30$
3. $TVC = 5 \times 500 = \text{£}2,500$; $TC = 2,000 + 2,500 = \text{£}4,500$; $AC = 4,500 \div 500 = \text{£}9$
4. $MC = (1,215 - 1,200) \div 1 = \text{£}15$
5. $TR = 12 \times 800 = \text{£}9,600$; $\text{profit} = 9,600 - 8,200 = \text{£}1,400$; $\text{margin} = 1,400 \div 9,600 \times 100 = 14.6\%$
6. $\% \Delta P = 2 \div 8 \times 100 = +25\%$; $\% \Delta Q = -75 \div 500 \times 100 = -15\%$; $PED = -15 \div 25 = -0.6$ (inelastic). TR rises from £4,000 to £4,250, because a price rise on inelastic demand raises revenue.
7. $(42,000 - 50,000) \div 50,000 \times 100 = -16\%$
8. Continue in the short run: AR (£6) is above AVC (£5.50), so it covers variable costs and contributes £0.50 per unit towards fixed costs. In the long run it should leave: AR (£6) is below AC (£7), so it makes a loss of £1 per unit and cannot cover all costs.

Table answers

Q	AR	TR	MR	TC	MC	AC	Profit
1	20	20	20	12	12	12.0	8
2	18	36	16	22	10	11.0	14
3	16	48	12	30	8	10.0	18
4	14	56	8	40	10	10.0	16
5	12	60	4	54	14	10.8	6
6	10	60	0	72	18	12.0	-12

9. **Q = 3, profit £18.** At 3 units MR (£12) is above MC (£8), so the third unit adds to profit. The fourth unit has MC (£10) above MR (£8), so it reduces profit. Profit is highest at 3 units.
10. **Q = 5 or 6** (TR is £60 at both). MR from the sixth unit is £0, so revenue stops rising at that point; revenue is at its maximum where MR reaches zero.
11. **Q = 5.** At 5 units AR (£12) is above AC (£10.80); at 6 units AR (£10) is below AC (£12), a loss. This is **sales maximisation** (output where $AR \approx AC$, with no loss).
12. Profit maximisation: price £16 at 3 units. Sales maximisation: price £12 at 5 units. Price is £4 lower (25% lower) under sales maximisation, and output is higher.